

Output Builder — designing a print layout

The Output Builder is where you draw a reusable **print template** — a care label, price sticker, carton marking, hang-tag — position text and graphics on a true-size page, and wire in the **variables** that fill it per style. This guide walks every control on the screen: what it means and how to use it.

Admins only · open it at **Output builder** ▶ click a layout (or **+ New**). The **?** beside **Save / Open PDF** opens this guide in a side panel while you work.

0 How a layout becomes a printed document

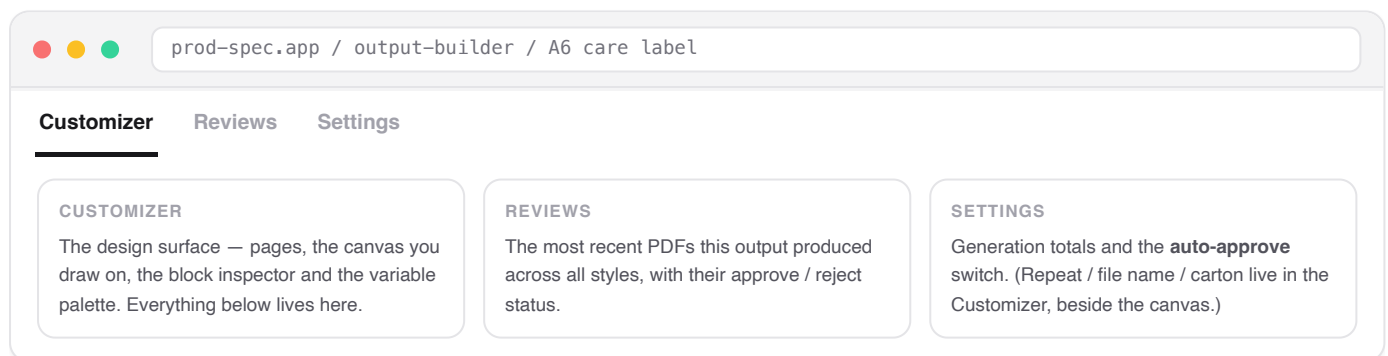
A layout on its own prints nothing. It becomes a real output only after you **Publish** it and add it to a style's **Prod Spec**. Keep this five-step path in mind — most of the builder's settings make sense once you see where the output ends up.

- 1 Design** the layout here — page size, blocks, variables — with a real test style in the live preview.
- 2 Publish** it. Publishing validates every variable and turns the Draft into a versioned, usable output `layout:<id>`.
- 3 Add it to a Prod Spec** (Prod specs ▶ open a spec ▶ add output). Only then is it attached to styles.
- 4 Generation** renders a PDF per style (and per size/EAN, if you set it to repeat), filling the variables from that style's data.
- 5 Review** — each PDF is approved or rejected, then sent to the supplier from the style page.

Once published, edits go live on save

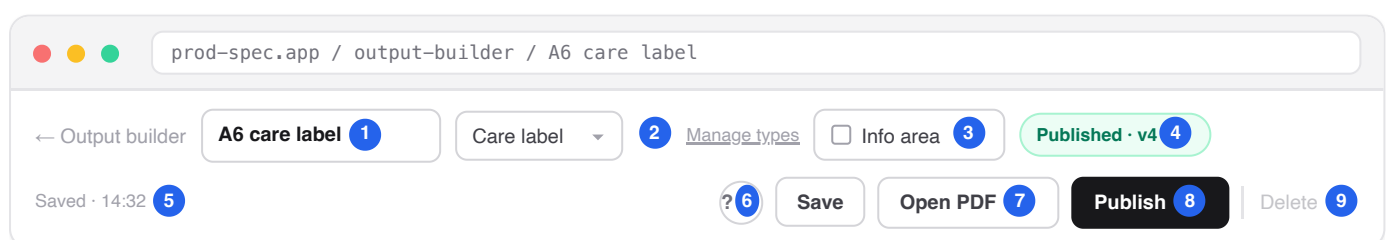
A **Published** layout is used by every spec that references it. Your next autosave changes what those specs generate from then on — there's no separate re-publish step for edits. Draft layouts are private until you publish.

The editor has **three tabs** — you'll spend almost all your time in the first:



1 The header — identity, saving & publishing

The sticky top bar carries the layout's identity on the left and its actions on the right. It's the same on every tab.

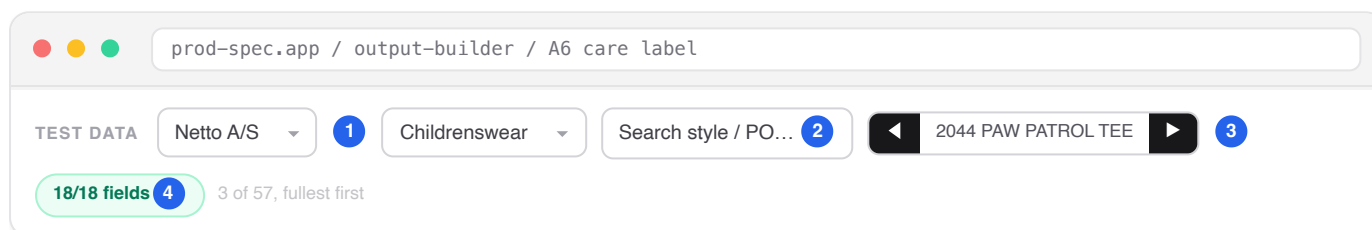


- 1 Name** — click to rename. Purely a label for you; it's how the layout shows in the list and the Prod Spec output picker.

- 2 **Document type** — groups the output in pickers and stamps every generated asset (e.g. **Care label**, **Carton marking**). Some app behaviour keys off it (carton outputs get extra powers). **Manage types** opens the catalogue to add / rename a type and set keyword **exclusion rules**.
- 3 **Info area** — tick this and the layout's print size becomes **switchable per style** on the Style page (the admin size, or a custom one). For fixed-size labels leave it off. Sizes are managed under **Settings** → **Info area sizes**.
- 4 **Status pill** — Draft (private, not usable yet) or Published · v# (live; edits go live on save).
- 5 **Save state** — *Unsaved changes* → *Saving...* → *Saved · time*. **Autosave** fires 1.2 s after your last change; the browser also warns before you leave with anything unsaved.
- 6 **?** — this guide, in a side panel you can read while you build (with a **Pop out** ↗ to the full page + PDF).
- 7 **Open PDF** — renders the **real** production PDF for the selected test style and opens it in a new tab. The truest proof of what will print.
- 8 **Publish** (Draft only) — validates the whole layout and makes it usable. If a variable is malformed or a required argument is missing, publishing is blocked and the reasons are listed here.
- 9 **Delete** — removes the layout and unlinks it from any Prod Spec (it tells you which). **Already-generated PDFs are kept** — only the link is removed.

2 Test data — preview with a real style

Pick a **customer** and **business area**; the builder finds real styles for that pair and previews your layout against one. This only drives the **preview** — it has no effect on what the output generates in production.



- 1 **Customer × Business area** — the pair that scopes which styles you can preview with. Both are required before styles load.
- 2 **Search** — jump to a specific style number or PO instead of cycling.
- 3 **◀ Style ▶** — step through the matches, ranked **fullest first** (the styles that fill the most of your variables come first, so you preview against complete data).
- 4 **Fields badge** — how many of the layout's variables this style fills. 18/18 is complete; an amber badge lists what's missing, so you can see gaps before they reach a real label.

Your test style now stays put

Editing the layout re-ranks the fullest-first list in the background, but the style you're previewing **stays selected** — the preview no longer jumps back to the first style each time. Changing customer, business area or the search is what re-picks a style.

3 Pages, page setup & the grid

The left rail defines the **paper**. A layout can have several pages (e.g. a front and

a back); each has its own size, margins and placement grid.

PAGES1

1 - Front
105 × 148 mm

2 - Back
105 × 148 mm

+ Add page

PAGE SETTINGS

TITLE
Front

WIDTH MM105

HEIGHT MM148

MARGINS MM2LINKED
4

ORIENTATION3
PortraitLandscape

GRID4
CELL SIZE (MM)
4

Regenerate

Current grid 26 × 37 · ≈ 4.0 × 4.0 mm per cell.

- 1 Pages — click to switch; + Add page clones the last page's size. Hover a page to reveal its × (a page with blocks asks first). The canvas always shows the selected page.
- 2 Margins — an inset from the paper edge that the whole grid (and every block) sits inside. LINKED drives all four sides from one value; click it to switch to per side (Top / Right / Bottom / Left). Type decimals with a comma or dot.
- 3 Orientation — Portrait / Landscape simply swaps width and height.
- 4 Grid — blocks snap to a grid of cells. Set a cell size in mm and press Regenerate to recompute the columns × rows; existing blocks are remapped proportionally (anything that no longer fits is nudged to fit). Smaller cells = finer placement.

Print guides — sewing & folding lines

Below the grid, add non-printing reference rules: a sewing line a fixed distance from the top or bottom edge (the seam allowance), and a folding line down or across the centre. They appear on the canvas and in the true render to help you place content, matching how the label is sewn or folded.

4 The canvas — drawing blocks

The centre pane is the page at true aspect ratio. You place content by drawing a block: press and drag a rectangle on the grid. Everything on the label is a block — a line of text, a barcode, a logo.

CANVAS — 105 × 148 MM · PORTRAIT

1

{{styleName}}

{{composition:da}}

[[barcode:ean13]]

ooo

Drag on the grid to draw a block · click a block to edit · Del removes the selected block.

BLOCKS24

{{styleName}}R2-C2

composition:da

BARCODE{{barcode:ean13}}

WASH{{washSymbols}}

- 1

A selected block shows a dark ring and an X. Click any block to select and edit it in the inspector (next section). Drag on empty grid to draw a new one.
- 2

Blocks list — every block on the page in reading order (top→bottom, left→right). Far easier than hunting the crammed canvas: click a row to select it, hover to flash its position with a blue locator ring, and use the row X to delete. Graphic blocks get a small kind tag (Barcode, Wash, Logo, Cert).

5

The block inspector — styling a block

With a block selected, the right rail shows everything about it: where it sits, how the text is styled, and — at the bottom — its **content**. All of it autosaves.

Layout controls

- **Column / Row / Width / Height** — the block's cell position and span. Type exact numbers when dragging isn't precise enough. The **≈ mm** readout shows its real printed size.
- **Align** (left / center / right) and **Vertical** (top / middle / bottom) — where the text sits inside the block.
- **Font size** in pt. **Barcodes and wash symbols scale with the font size** — bump the pt to enlarge a barcode.
- **Bold** and **Line height** (1.2–1.8).

Box & fit controls

- **Border** — width (0.2–2 mm), colour, and inner **padding** (linked or per side). A framed field.
- **Invert** — white text on a solid black block.
- **Fit width** — force the text onto one line and auto-scale it to fill the block's width.
- **Shrink to fit cell** — reduce the text until it fits its box, so it never overflows onto neighbours.

The content box — one line per printed row

Each line in the content box is one printed row. Type `{{` to open **variable autocomplete**; wrap a selection in `**bold**` or `_italic_` (the **B / I** buttons do this) to style part of a line. Literal text prints as-is; variables resolve at generation. What you type is exactly what the preview and the PDF render.

6

Variables — the heart of a layout

Variables (also called **tokens**) are `{{placeholders}}` that fill from the style's data at generation. The **Variables palette** on the right lists them by group; click a chip to insert it at your cursor. Tick **Show values** to resolve every chip against the current test style so you can see real data.

Text variables — the everyday fields. Insert as `{{styleNumber}}` etc.

GROUP	EXAMPLES (NOT EXHAUSTIVE)
Style	{{styleName}} {{styleNumber}} {{description}} {{customerName}} {{customerItemNo}} {{colourName}} {{sizes}} {{size}} {{sizeRange}} {{price}} {{countryOfOrigin}} {{certificates}}
Order & carton	{{poNumber}} {{customerOrderNo}} {{qtyPerCarton}} {{cartonEan}} {{assortEan}} {{ean13}} {{batchNo}} {{deliveryTerm}} {{supplierNumber}}
Per language need a <code>:lang</code>	{{composition:da}} {{careInstructions:en}} {{productName:de}} {{madeIn:da}} {{country:fr}} — pick the language from the palette's Per language dropdown so the right variant inserts.

Graphics — rendered as images, scaled by the block's font size.

Barcodes	{{barcode:cartonEan}} carton EAN as Code 128; {{barcode:cartonEan13}} the same carton number as a true EAN-13; {{barcode:ean13}} the first size's product EAN-13. Add a bar height in mm: {{barcode:ean13:8}} (2–40 mm).
Wash symbols	{{washSymbols}} the style's care icons as a row. Optional mm gap between icons: {{washSymbols:0}} (0 = flush).

Logos	{{logo:contrast}} / {{logo:contrastAddress}} the built-in Contrast marks; {{logo:custom}} this layout's own uploaded logo (see §7).
Certificate marks	{{cert:oekotex}} {{cert:fsc}} — self-gating : a mark prints only on styles whose Certificates field declares it. No {{if}} wrapper needed. Artwork comes from Settings → Certificates (a placeholder shows until the licensed mark is uploaded).

Logic —

{{if ...}}{{else}}{{endif}}

Conditional text, **one condition per line, no nesting**. The condition field must be a text variable.

- == / !=

 compare the whole value, case-insensitively:

{{if deliveryTerm == FOB}}...{{else}}...{{endif}}
- includes / !includes

 test a comma-separated list:

{{if certificates includes FSC}}FSC certified{{endif}}

Calculated —

{{= ... }}

Arithmetic over field values, useful on multi-style cartons.

- + - × ÷

 on numeric fields:

{{= qtyPerCarton * 2 }}
- sum / count / min / max(field)

 across the styles on the carton (base + picked siblings; an empty slot counts as 0):

{{= sum(qtyPerCarton) }}
- round(..., decimals)

 to tidy the result.

Sibling styles — putting several styles on one carton

When **Multiple styles on the box** is on (§7), a carton layout can place OTHER styles from the same PO via slot tokens:

{{style2}}

 (the headline number) and its fields

{{style2Name}}

,

{{style2Description}}

,

{{style2QtyPerCarton}}

 ... up to 8 slots. Pick the slot from the palette's **Sibling styles** dropdown. Branch the single- vs multi-style look with

{{if multipleStyles == true}}...{{else}}...{{endif}}

 — note

==

, not

===

.

On the canvas and in the content box, variables are colour-coded:

known

 muted,

{{if}}/{{else}}

 italic,

{{= calc}}

 blue (red if the expression is invalid), and

{{unknown}}

 red — an at-a-glance check before you publish.

7 Output settings — repeat, split, file name, carton & logo

Below the grid in the Customizer's left rail sit the settings that decide **how many documents** this output makes per style and **what they're called**. This is where most of the "possibilities" live.

SETTINGS

REPEAT OUTPUT 1

Per EAN (size × colour)

▼

3 repetitions: 98/104-Navy, 110/116-Navy, ...

OUTPUT FILES 2

One PDF per EAN

▼

OUTPUT FILE NAME 3

{{styleNumber}}-{{size}}-care

→ IL97261-98_104-care

☐

Carton numbering (X/Y) 4

☐

Multiple styles on the box 5

1

Repeat output

— make one document, or one per variant: **Don't repeat** (a single PDF) · **Per size** · **Per EAN** (size × colour) · **Per assortment EAN** · **Per carton EAN** (per size + assort). The panel shows the exact repetitions for the test style. Use

{{size}}

 /

{{ean13}}

 in the design so each repetition shows its own data.

2

Output files

(only when repeating) — **One PDF per EAN** (a separate file for each repetition, each holding only its own EAN) or **One single PDF** (every repetition bound into one file).

- 3 **Output file name** — a template for the generated file's name. Variables are allowed (`{{styleNumber}}` , `{{size}}` , `{{colourName}}` ...); the resolved name previews live in green. Leave it empty for the default name. When splitting per EAN, include a per-EAN variable so each file gets a distinct name.
- 4 **Carton numbering (X/Y)** — lets an operator print a numbered set (1/N ... N/N) from the Style page. Turn it on, then place `{{cartonNo}}` / `{{cartonTotal}}` (or zero-padded `{{cartonNoPadded}}`) on the label. The builder warns if the switch is on but the tokens aren't placed, and the preview gains a “**Preview as carton N of M**” control.
- 5 **Multiple styles on the box** — enables the sibling slots from §6 so an operator can add other same-PO styles to a carton as a manual one-off. Independent of carton numbering.

Custom logo

The moment your design uses `{{logo:custom}}` , a **Custom logo** control appears here: upload an SVG / PNG / JPG (under ~450 KB) for **this layout**, and set its printed **width as a % of its block** (height keeps the aspect ratio). It's stored per layout, separate from the autosave.

8 The live preview — true render

Under the canvas, the **Print preview** is rendered by the **same code that generates the production PDF** — not an approximation. It updates a moment after each change.

PRINT PREVIEW — TRUE RENDER · 2044 PAW PATROL TEE
1 unresolved: `{{batchNo}}` 2

2044 PAW PATROL TEE 1

CARE · PREVIEW

`{{batchNo}}`

■■■■■

- 1 **Real data** from the selected test style fills the variables. With no style picked it renders with **sample data** and says so.
- 2 **Unresolved variables** show as **amber chips** here and are counted in the corner. They **never print** — an amber chip means that style has no value for that field, so fix the data or guard the line with `{{if}}` . When everything resolves you get a green “*all variables resolved*”.

Edit as JSON

At the bottom of the right rail, **Edit JSON directly** exposes the whole layout — pages, margins, blocks, repeat / split / file-name settings — as one JSON document. An escape hatch for bulk tweaks and copy-paste between layouts; it's validated before it applies.

9 The Settings & Reviews tabs

Settings tab

Generation activity — how many PDFs this output has ever produced (total / approved / pending / rejected), across how many styles, and when it last ran.

Auto-approve outputs — a switch that marks this output's assets **approved the moment they generate**, skipping the manual review queue. Two guards stay: a print-unsafe document (missing artwork or EAN) still falls back to manual review, and a person still presses “**Approve all & publish**” on the style to send to the supplier.

Reviews tab

The most recent PDFs this output produced, newest first — style, file name (with a placeholder-count flag), **status** (Pending / Approved / Rejected), when it generated, and a link to **Open PDF**. The tab shows an amber count when any are awaiting review.

Open a style's review screen from here to actually approve or reject.

In one line: draw blocks on the true-size grid, fill them with `{{variables}}` from the palette, watch the true-render preview against a real style, set repeat / file-name / carton options, then **Publish** and add the output to a Prod Spec so it generates.

